The Re-prompt Tax Persists: Measuring Hidden Interaction Costs for Multilingual and Code-Switched LLM Users

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Paper under double-blind review

Abstract

Multilingual users often interact with large language models (LLMs) through code-switched prompts, mixed-language editing requests, and constraints about script, register, locale, and literal text preservation. Standard one-turn evaluations usually ask whether the final answer is correct, but they miss a common usability failure: the model returns a fluent, plausible answer that violates the user's interaction contract, forcing the user to spend extra turns repairing the response. We introduce *Re-prompt Tax*, a metric family that measures first-turn global alignment, the number of repair turns needed for recovery, unresolved failures, and token overhead. We construct REPROMPTTAX-STRESS-V0.2, a 120-item synthetic stress pilot across Spanish-English, Hindi-English, and Arabic-English interactions, and evaluate three GPT-4.1-family API models under a baseline prompt and a Global Interaction Contract prompt. Baseline first-turn global alignment ranges from 67.5% to 76.7%; the mitigation improves alignment to 76.7–93.3% and reduces mean repair turns and repair-normalized token tax. A current-model refresh shows the same pattern on full 120-item `gpt-5.4-mini` and `gpt-5.5` runs: under `gpt-5.5`, alignment rises from 81.7% to 98.3%. Prompt-mechanism diagnostics show that a narrower content-preservation scaffold captures most of this gain, but only near-ties the full contract on current models. Analysis shows that the dominant failure mode is not generic multilingual QA, but implicit editing preservation: models often translate or frame content in the instruction language when the user intended the quoted content to remain in English. A 10% stratified blinded LLM-judge audit agrees with the automatic scorer on 71/72 sampled pass/fail labels, and a `gpt-5.5` judge refresh agrees on 70/72. We release a conservative evaluation protocol and claim boundary for measuring hidden global usability costs rather than broad multilingual ability.

1 Introduction

The GenAI4World workshop asks how generative AI systems should be evaluated for global tasks, contexts, vulnerabilities, and usability beyond standard accuracy metrics (GenAI4World Organizers, 2026). This question matters because many real users do not interact with LLMs through clean monolingual benchmark items. A single request may contain an instruction in one language, content in another, an English file name embedded in Hindi written in Latin script, or a local date, currency, or honorific convention. A model may understand the literal task and still violate the pragmatic contract that makes the response usable.

Consider the following Spanish-English editing request:

```
Me ayudas a mejorar este texto para que suene natural? ‘‘I can’t join the
call today, but I can share my update by email.’’
```

A globally aligned assistant should infer that Spanish is the instruction language and that the quoted English sentence is the object of editing. Unless translation is requested, the edited sentence should remain English. A response that translates the sentence into Spanish is fluent and superficially helpful, but it is unusable for the user's intended workflow. The user must diagnose the failure, issue a corrective follow-up such as "Please keep the rewritten text in English," and check whether the repair succeeded. We call this extra burden the *Re-prompt Tax*.

Existing multilingual evaluations have greatly expanded language coverage and have shown that translated benchmarks can be culturally distorted (Singh et al., 2024; Xuan et al., 2025; Huang et al., 2025b). Other work has studied multilingual instruction following (Zhou et al., 2023; Li et al., 2025) and output-language alignment in code-switched interactions (Oh et al., 2026). These efforts are essential, but they leave an interactional gap. A user-facing system is not only a question-answering engine; it is a collaborator in a multi-turn workflow. When a first response violates a language, script, preservation, register, or locale constraint, the cost is paid by the user in extra turns, extra tokens, and uncertainty about whether the model will repair the failure.

This paper treats multilingual usability as an *interaction contract* between a user and a model. The contract may specify, explicitly or implicitly: (1) the task, (2) the expected response language, (3) the language of content to be edited or summarized, (4) script or romanization requirements, (5) literal spans that must not be changed, and (6) register or locale constraints. A model can violate this contract even when a semantic task label such as "rewrite" or "summarize" is satisfied. Measuring only one-turn task success therefore understates the burden imposed on multilingual and code-switched users.

Our contributions are:

- We define **Re-prompt Tax**, a metric family for hidden multilingual repair burden: first-turn global alignment, re-prompt turn tax, Repair@1/2, unresolved rate, and repair-normalized token tax.
- We construct REPROMPTTAX-STRESS-V0.2, a balanced 120-item synthetic stress pilot over three language pairs and four task families: implicit editing preservation, output-language inference, quote preservation, and script/register/locale constraints.
- We evaluate three GPT-4.1-family API models (OpenAI, 2025) and full 120-item current-model refreshes for `gpt-5.4-mini` and `gpt-5.5`; under `gpt-5.5`, the contract improves first-turn global alignment from 81.7% to 98.3% while leaving inspectable residual failures.
- We provide diagnostics beyond the headline table: language slices, family-level failure modes, paired sign-test sensitivity, token-burden analysis, prompt-mechanism and repair-realism diagnostics, item-consistency analysis, paired judge refreshes, and launch-ready human/native review protocols with conservative claim gates.

We intentionally keep the claim boundary narrow. The pilot is synthetic and stress-oriented, not a prevalence estimate for all global users. Its purpose is to make a neglected class of repair costs measurable and reproducible.

2 Interaction contracts and metrics

Interaction contract. For each benchmark item i, we specify a user prompt u_i and a contract c_i. The contract records expected response language, expected script, required task behavior, literal spans that must be preserved, forbidden markers or translations, and any register or locale constraints. Some contract elements are explicit in the prompt, while others are implicit but recoverable from the workflow. In the Spanish-English editing example above, the prompt asks for text improvement in Spanish, but the target content is quoted in English. The contract therefore requires English edited output, not Spanish translation.

First-turn global alignment. For model m and condition s, we define first-turn global alignment as:

$$\mathrm{FTGA}(i, m, s) = \begin{cases} 1 & \text{if the first response satisfies all contract checks,} \\ 0 & \text{otherwise.} \end{cases}$$

Family	User-side ambiguity	Expected contract
Editing preservation	Instruction language differs from quoted content language.	Improve or rewrite the content while preserving its intended language; do not translate unless asked.
Output-language inference	Prompt mixes languages without an explicit "answer in X" instruction.	Infer the expected output language from the pragmatic context.
Quote preservation	The task concerns text around quoted headings or labels.	Preserve required quoted spans and literal data exactly.
Script/register/locale	The user specifies romanization, formal/informal register, or local literal formats.	Obey script, register, and locale constraints while preserving IDs, file names, times, and headings.

Table 1: The four task families in REPROMPTTAX-STRESS-V0.2. The pilot is designed to stress interaction contracts, not to estimate natural prevalence.

The checks cover language, script, preservation, task completion, and register/locale constraints. FTGA is stricter than ordinary task success: a response can complete the requested operation while still failing because it uses the wrong response language, translates a quoted span, or changes a literal file name.

Repair trajectory. If the first response fails, we append up to two standardized repair prompts associated with the item. The repair prompts point to the violated contract class without revealing a reference answer. The **Re-prompt Turn Tax** (RTT) is the minimum number of repair prompts required before success:

$$\mathrm{RTT}(i, m, s) \in \{0, 1, 2, 3\},$$

where 0 denotes first-turn success and 3 denotes unresolved after two repair prompts. We also report Repair@1, Repair@2, and unresolved rate among initially failed trajectories.

Token tax. Let $T_{\mathrm{total}}(i, m, s)$ be provider-reported total tokens consumed by the trajectory through success or exhaustion of the two-repair budget, and let $T_0(i, m, s)$ be the first-attempt token count. We define:

$$\mathrm{TokenTax}(i, m, s) = \frac{T_{\mathrm{total}}(i, m, s)}{T_0(i, m, s)}.$$

Token tax is a repair-burden ratio, not an API-cost estimate. A longer system prompt can reduce repair-normalized token tax while increasing absolute tokens. We therefore report absolute token burden separately in the analysis.

3 Benchmark construction

Motivation from public chat logs. As a lightweight motivation check, we scan the first 20,000 streamed WildChat conversations (Zhao et al., 2024) using predefined repair cues. Among 10,681 multi-turn conversations, 172 contain at least one follow-up cue related to generic repair, preservation, output language, unwanted translation, script, or register/locale. The scan records only aggregate counts and hashed metadata: no raw user or assistant text is written. We use the scan to motivate the failure taxonomy, not to estimate population prevalence.

Stress-pilot design. We created REPROMPTTAX-STRESS-V0.2, a 120-item synthetic benchmark with a balanced 3×4 design: Spanish-English, Hindi-English, and Arabic-English; and four task families with ten items per language-pair/family cell. Table 1 summarizes the families.

The benchmark is synthetic and privacy-safe. A release-hygiene audit confirms 120 unique prompts, zero normalized duplicate prompts, required scoring markers for all rows, zero privacy-marker hits under the audit regexes, and 99 literal preservation spans across the families that require them. The benchmark also contains known-bad-output notes to clarify what each item is designed to catch. These checks are not a substitute for native-speaker validation, but they make the artifact easier to audit and reproduce.

4 Experimental protocol

Models. We evaluate `gpt-4.1-nano`, `gpt-4.1-mini`, and `gpt-4.1`, all accessed through the API with temperature 0 and a maximum of 256 output tokens. The original paper-facing experiments were run on June 28, 2026. Because OpenAI's current model documentation lists GPT-5.5 as the flagship model and GPT-5.4 mini/nano as lower-cost variants (OpenAI, 2026), we add a current-model refresh on full 120-item `gpt-5.4-mini` and `gpt-5.5` runs using the same benchmark, conditions, scoring rules, and repair budget.

Prompting conditions. Each model is run under two system-prompt conditions. The baseline prompt is "You are a helpful assistant." The mitigation condition uses a **Global Interaction Contract** prompt that instructs the model to infer instruction and content language separately, preserve content language during editing unless translation is requested, preserve quoted spans and literal data, obey explicit script/register/locale constraints, and avoid unnecessary explanatory framing.

Repairs. For each failed first response, we append the item's first standardized repair prompt and rescore the next response. If the response still fails, we append a second standardized repair. The trajectory stops at first success or after two repairs. This fixed repair budget lets us compare models and prompts under the same user burden. It also separates first-turn usability from recoverability: some models fail initially but repair quickly, while others continue to violate literal-preservation constraints.

Scoring. Automatic scoring combines deterministic checks for exact span preservation and script with rule-based language and task checks. The scorer emits both a binary FTGA label and component labels for language, script, preservation, task, and register/locale. We run scorer-ablation diagnostics by relaxing one component at a time, and deterministic regression tests cover controlled language, script, preservation, task, and register/locale failures.

Audit. We run a blinded `gpt-4.1` judge audit on 72 first-turn responses, sampling three responses from each model-condition-family stratum. The judge sees the user prompt, expected contract, and assistant response, but not model identity or prompting condition. This is a scorer sanity check, not a native- or near-native human validation study.

5 Results

5.1 Main trajectory metrics

Table 2 shows measurable baseline Re-prompt Tax for the GPT-4.1-family rows and the GPT-5.x refresh rows. The Global Interaction Contract improves FTGA for every model, with GPT-4.1-family gains of +9.2 points for nano, +3.3 points for mini, and +16.7 points for GPT-4.1. Mean RTT also falls for the GPT-4.1-family rows. The strongest GPT-4.1-family model reaches 93.3% FTGA under the contract, but the unresolved rate does not vanish; residual failures are concentrated in exact-preservation and script/register/ locale cases.

Paired sign-test sensitivity supports the strongest mitigation effects while clarifying the weaker mini result. For GPT-4.1, the contract improves FTGA on 20 paired items, worsens none, and ties on 100 items ($p < 0.0001$); mean RTT falls by 0.133 and mean token tax by 0.300. For nano, FTGA improves on 12 items, worsens on 1, and ties on 107 ($p = 0.0034$); mean RTT falls by 0.142 and token tax by 0.351. Mini's FTGA movement is smaller: 4

Model	Condition	FTGA	Mean RTT	Tok. tax	Unres.	R@1	R@2
GPT-4.1 nano	Baseline	67.5	0.47	1.69	5.8	74.4	82.1
GPT-4.1 nano	Contract	76.7	0.33	1.34	4.2	78.6	82.1
GPT-4.1 mini	Baseline	75.8	0.28	1.43	0.8	89.7	96.6
GPT-4.1 mini	Contract	79.2	0.24	1.27	1.7	92.0	92.0
GPT-4.1	Baseline	76.7	0.28	1.43	2.5	89.3	89.3
GPT-4.1	Contract	93.3	0.15	1.13	4.2	37.5	37.5
GPT-5.4 mini	Baseline	80.0	0.25	1.38	2.5	87.5	87.5
GPT-5.4 mini	Contract	85.0	0.25	1.24	5.0	66.7	66.7
GPT-5.5	Baseline	81.7	0.23	1.28	1.7	86.4	90.9
GPT-5.5	Contract	98.3	0.02	1.02	0.0	100.0	100.0

Table 2: Stress-pilot results, including the current-model refresh rows. FTGA, unresolved rate, Repair@1, and Repair@2 are percentages. RTT and token tax are lower-is-better; token tax is total tokens until success or exhaustion divided by first-attempt tokens.

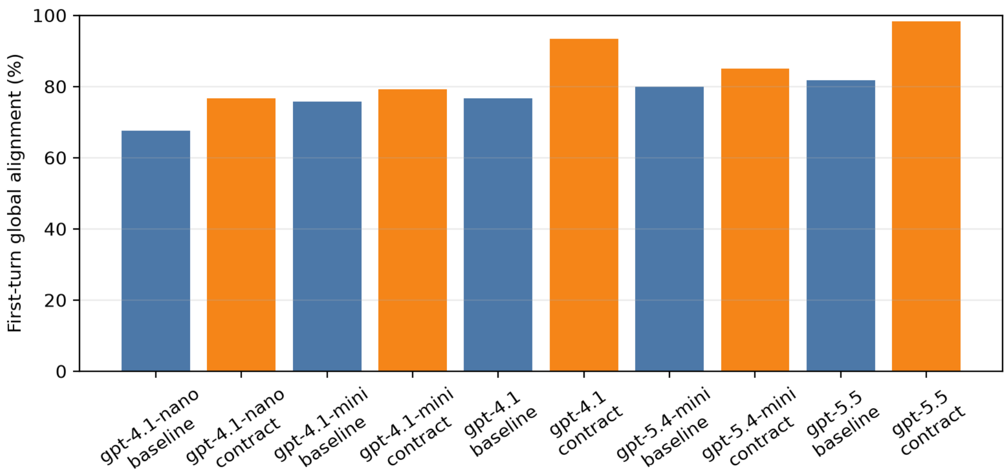

Figure 1: First-turn global alignment by model and condition on the 120-item stress pilot, including the GPT-5.x current-model refresh rows.

improved, 0 worsened, and 116 tied ($p = 0.125$), although token-tax reduction is consistent at 0.16.

The current-model refresh strengthens the timeliness claim. On the same 120 items, `gpt-5.4-mini` moves from 80.0% to 85.0% FTGA with lower token tax but a higher unresolved rate; its +5.0 point item-bootstrap interval crosses zero at [-0.8, 11.7]. The full `gpt-5.5` run is clearer: FTGA rises from 81.7% to 98.3% (+16.7 points, [10.0, 24.2], 20 improved/0 worsened), mean RTT falls from 0.225 to 0.017, and unresolved trajectories fall to 0.0% (Table 2). The contract still leaves two GPT-5.5 first-turn failures, both Spanish-English editing-preservation rows that repair in one turn; the lower-cost model has 11 fixes, five first-turn regressions, and six unresolved contract trajectories. A generation-progress probe shows the same pattern at item level: GPT-5.x-family baseline rows reduce the normalized failure-pair rate from 96/360 for GPT-4.1-family baselines to 46/240, and `gpt-5.5` with the contract passes 38 of the 40 items where at least one GPT-4.1-family baseline failed.

5.2 Failure families

The main burden comes from implicit editing preservation, not from generic multilingual output-language inference. Table 3 shows that aggregate baseline FTGA is 33.3% for editing preservation, with 60/90 first-turn failures across models. In contrast, output-language

Task family	Baseline FTGA	Contract FTGA	Baseline fail.	Contract fail.
Editing preservation	33.3	70.0	60/90	27/90
Output-language inference	100.0	100.0	0/90	0/90
Quote preservation	90.0	91.1	9/90	8/90
Script/register/locale	70.0	71.1	27/90	26/90

Table 3: Family-level first-turn alignment aggregated across the three models. Editing preservation is the dominant source of baseline burden; output-language inference is saturated in this pilot.

inference is saturated at 100.0% FTGA in this particular pilot. The contract improves editing-preservation FTGA to 70.0%, reducing mean RTT for that family from 0.667 to 0.300. The family-level effect is largest for `gpt-4.1` (+63.3 points on editing preservation), followed by mini (+13.3) and nano (+33.3); in model order, the deltas are +63.3, +13.3, and +33.3.

The component breakdown explains why editing preservation is difficult. In baseline editing-preservation rows, the first-turn language check passes only 33.3% of the time and the script check passes 66.7%. Under the contract, those rates rise to 70.0% and 92.2%, respectively. However, preservation pass rates barely move in some other families, and script/register/locale preservation remains below saturation. This suggests that the contract helps models separate instruction language from content language, but it does not fully solve literal span preservation. For nano quote-preservation rows, the contract lowers mean RTT from 0.63 to 0.37 and unresolved rate from 13.3% to 3.3%, but first-turn exact preservation remains a capability boundary.

5.3 Language slices and item consistency

The aggregate gain is not evenly distributed across language pairs. Averaged across models, FTGA increases by +20.0 points on Arabic-English, +9.2 points on Spanish-English, and +0.0 points on Hindi-English. This does not mean that Hindi-English is easier in general; in this stress pilot, the Hindi-English slice has little room to improve because baseline FTGA is already high for all three models. The strongest language-slice improvements are GPT-4.1 on Arabic-English and Spanish-English (+25.0 points each) and nano on Arabic-English (+25.0 points). The weakest slice is Spanish-English on `gpt-4.1-mini`, where FTGA delta is +0.0 points despite lower token tax.

Item-consistency analysis separates systematic hard items from scattered one-model failures. Under baseline prompting, 40/120 items fail for at least one model and 27/120 fail for all three. Under the contract, 35/120 items still fail for at least one model, but only 8/120 fail for all three. The contract reduces the number of failing models on 22/120 items and increases it on only 1/120 item. The hardest residual items are concentrated in script/register/locale rows requiring exact preservation of local literal spans.

5.4 Repair dynamics and token burden

Repair trajectories reveal a distinction between first-turn usability and recoverability. For GPT-4.1, the contract changes the distribution from 92 first-turn successes and 28 initial failures to 112 first-turn successes and 8 initial failures. For nano, first-turn successes rise from 81 to 92 and unresolved trajectories fall from 7 to 5. Mini has a deliberately modest trajectory shift: 91 to 95 first-turn successes, with 5 paired RTT improvements, 1 worsening, and 114 ties.

Token tax decreases for all three models, but absolute token burden increases because the contract prompt is longer. For GPT-4.1, mean total tokens rise from 132.7 under baseline to 256.3 under the contract, even though mean extra tokens after the first turn decrease from 51.2 to 32.9 and token tax falls from 1.43× to 1.13×. We therefore interpret token tax as normalized repair burden rather than cost savings; the same caveat holds across all five full-run models, including GPT-5.5, where 36.8 fewer post-first-turn repair tokens accompany 114.5 more total tokens per item.

A 24-item repair-realism diagnostic shows that RTT is also sensitive to the wording of the repair. On baseline editing-preservation failures, the saved standard repair and a terse user-like repair both recover 24/24 sampled cases, but a frustrated repair recovers 17/24 and an explicit source-language contract recovers only 5/24, often because models keep explanatory framing in Spanish or Arabic. We treat this as interaction-realism evidence, not as a user study.

5.5 Prompt controls and mechanism evidence

We also run single-model prompt diagnostics on `gpt-4.1-nano`. A longer generic-helpfulness prompt improves FTGA from 67.5% to 75.0%, suggesting that some mitigation benefit comes from generic instruction-following scaffolding. The Global Interaction Contract reaches 76.7% FTGA and the lowest token tax in that three-condition control. A narrower content-preservation prompt performs best in a later ablation: 80.0% FTGA, 0.27 mean RTT, and 1.28$\times$ token tax. This is an important claim boundary. The three-model paper-facing intervention is the full Global Interaction Contract, but the best nano prompt tested is the narrower content-preservation scaffold. The result suggests that content language and literal-preservation rules drive much of the mitigation. On current models, the narrower prompt is a near tie rather than a clear winner: 85.8% FTGA versus 85.0% for the full contract on `gpt-5.4-mini`, and 99.2% versus 98.3% on `gpt-5.5`. The current-model gaps are only one net item, so we treat preservation scaffolding as mechanism evidence, not as a uniformly better prompt.

5.6 Scorer sensitivity and audit

Scorer-ablation diagnostics show that the headline trend is not caused by one fragile automatic rule. Relaxing language alone would move aggregate baseline FTGA from 73.3% to 76.4%; relaxing preservation alone would move it to 78.9%; relaxing task alone would not change it. Under the contract, relaxing language would move FTGA from 83.1% to 83.3%, and relaxing preservation would move it to 88.3%. Many failures violate multiple checks at once, especially editing failures that combine language, script, and task violations. The task-useful diagnostic separates task noncompletion from hidden repair burden: 31/96 baseline first-turn failures still pass the task component, and the stricter task-plus-preservation slice falls from 11 to 1 under the contract.

The blinded judge audit agrees with the corrected automatic scorer on 71/72 sampled pass/fail labels (98.6%; Wilson 95% CI 92.5–99.8). The single pass/fail disagreement is an editing-preservation response where the automatic scorer rejects Spanish framing around English rewrites while the judge accepts the answer. A paired `gpt-5.5` judge refresh on the same 72 rows agrees with the automatic scorer on 70/72 labels and with the original `gpt-4.1` judge on 69/72 labels. The stronger judge is stricter on the Spanish-framing case and flags two additional auto-pass rows for register or concision issues. Component agreement is lower for preservation and register/locale, which reinforces our conservative claim: LLM judges are useful as scorer sanity checks, but native/near-native validation remains necessary before making stronger cultural or register claims.

6 Discussion

Re-prompt Tax changes the unit of analysis from a single answer to a repair trajectory. This matters because a first answer can be semantically plausible but interactionally wrong. For global users, the cost of that error includes not only extra tokens, but also the cognitive work of diagnosing the failure, writing a corrective prompt, and verifying the repair. These costs are easy to miss when benchmarks ask only whether a final response is acceptable.

The strongest empirical finding is that implicit content-language preservation is a measurable bottleneck. Models often follow the language of the instruction rather than the language of the quoted content to be edited. A simple contract prompt reduces this failure, but does not eliminate exact-preservation errors. This pattern suggests that multilingual system prompts should separate *instruction language*, *content language*, and *response language* as distinct variables rather than treating the prompt as a monolithic language signal.

The results also show why mitigation claims should be bounded. The full Global Interaction Contract improves all three models, but it should be read as a pre-registered intervention, not as the best possible prompt: a narrower content-preservation prompt outperforms it on the nano diagnostic and near-ties it on current `gpt-5.4-mini` and `gpt-5.5` diagnostics. This suggests that more specific scaffolds may be better than broad multilingual policy text for some workflows, but the current-model gaps are too small to claim prompt dominance. Future systems could infer a lightweight interaction contract from the user's prompt and attach only the relevant constraints, reducing both absolute prompt length and over-instruction risk.

7 Related work

Multilingual and cultural evaluation. Multilingual benchmark work has expanded language coverage and highlighted large performance gaps across languages. MMLU-ProX evaluates advanced reasoning across typologically diverse languages (Xuan et al., 2025), BenchMAX proposes a broad multilingual evaluation suite (Huang et al., 2025b), and OMGEval evaluates open-ended multilingual generation with localized questions (Liu et al., 2024). Global MMLU shows that simply translating English benchmarks can preserve Western-centric cultural assumptions and distort model rankings (Singh et al., 2024). Cultural benchmarks such as CDEval and MCEval argue that model behavior should be evaluated through culturally specific and cross-cultural lenses (Wang et al., 2023; Huang et al., 2025a). Our work is smaller and narrower: it does not attempt broad cultural coverage, but instead measures the interactional cost of recovery after violating multilingual interaction contracts.

Instruction following and code-switching. IFEval introduced verifiable instruction-following checks for LLMs (Zhou et al., 2023); XIFBench and Marco-Bench-MIF extend this concern to multilingual instruction following with localized constraints (Li et al., 2025; Zeng et al., 2025). OLA is the closest prior work to ours: it studies output-language alignment in code-switched interactions and shows that models often infer the wrong response language (Oh et al., 2026). CodeMixQA and related work further show that code-switched reasoning and generation remain challenging (Winata et al., 2026). Re-prompt Tax includes output-language alignment, but adds repair trajectories and a broader contract containing script, preservation, register, and locale constraints, and standardized repairs are needed to compare those failures as trajectories rather than isolated first-turn answers.

Multi-turn dialogue, real-world chat logs, and judges. MT-Bench and Chatbot Arena evaluate conversational quality and preference using multi-turn prompts and LLM or human judgments (Zheng et al., 2023b). WildChat and LMSYS-Chat-1M provide large-scale public records of real LLM interactions (Zhao et al., 2024; Zheng et al., 2023a). We use WildChat only for a bounded aggregate cue scan and release synthetic stress items instead of raw user conversations. Our judge audit is related to LLM-as-a-judge methodology, but multilingual judging itself is known to be inconsistent across languages (Fu & Liu, 2025). We therefore treat the judge audit as a sanity check rather than as final human validation.

Cost and accessibility. Prior work on multilingual tokenization shows that language users can face unequal token costs and utility when using commercial APIs (Ahia et al., 2023; Petrov et al., 2023). Re-prompt Tax is complementary: instead of asking whether the same content costs more to encode in different languages, we ask how many extra turns and tokens are consumed when the model violates a multilingual interaction contract.

8 Limitations and ethics

This is a pilot benchmark, not a representative sample of multilingual or code-switched use. It covers only three language pairs and uses synthetic prompts. The English content embedded in the items reflects a common editing workflow, but many global workflows involve non-English target content, non-Latin scripts, dialectal variation, or richer social-register distinctions that are not covered here.

The automatic scorer is intentionally conservative and strongest for exact span-preservation and script checks. It is weaker for nuanced register and cultural appropriateness. Two blinded LLM-judge audits support most sampled pass/fail labels, but they also disagree on a small number of borderline editing, register, and concision cases. They are not a substitute for native/near-native human validation; native-speaker audit remains necessary before stronger final claims. Any release or final workshop version should clearly label the human audit status and avoid claims that the benchmark validates cultural appropriateness for all included communities.

The WildChat cue scan uses only aggregate counts and hashed metadata. We do not release raw user logs or quote private conversation text. The scan is useful for motivating the taxonomy, but regex cue hits are not reliable prevalence estimates. The benchmark items are synthetic and should be used to stress-test interaction failure modes, not to characterize all speakers. Any release should include a benchmark card explaining intended use, non-representativeness, language-pair coverage, and known scorer limitations.

Finally, token tax is a normalized repair-burden metric, not a direct economic claim. In our experiments, the longer contract prompt reduces repair-normalized token tax but increases absolute token use. Reporting both numbers is important for users whose constraints are monetary cost, latency, or context-window budget.

9 Conclusion

We introduced Re-prompt Tax as a way to measure hidden interaction costs for multilingual and code-switched LLM users. On a 120-item synthetic stress pilot, three GPT-4.1-family models exhibit measurable first-turn contract failures, especially on implicit editing-preservation tasks. A lightweight Global Interaction Contract prompt improves first-turn global alignment and reduces repair burden, and a full current-model refresh shows the same pressure on `gpt-5.5`: first-turn alignment improves from 81.7% to 98.3% but is not perfect. Follow-up prompt-mechanism diagnostics indicate that narrower content-preservation scaffolding captures much of the effect, but the full contract remains the paper-facing intervention. Residual exact-preservation and script/register/locale failures remain. The broader takeaway is methodological: global evaluation should measure not only whether a model can eventually produce a correct answer, but also how much repair work users must perform to make the answer usable.

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